



Individual Assignment

Project Overview:

You will choose a **real-world structural component**—such as a **crane arm**, **shelf bracket**, **handrail**, **machine frame beam**, **bicycle frame member**, or **bridge girder**—and investigate its behavior under approximate loading conditions.

You will:

- Identify loading types and support conditions.
- Determine shear forces, bending moments, and critical stresses.
- Consider varying cross-sections or material changes.
- Evaluate the component's performance and safety.

Objective:

- Apply theory to practical engineering structures.
- Analyze structural components with varying cross-sections.
- Construct shear force and bending moment diagrams.
- Evaluate stress distribution and factor of safety.
- Explore design trade-offs (shape, material, size).

Tasks:

1. Component Selection & Description
2. Load & Support Assumptions
3. Analytical Investigation
4. Cross-Sectional & Stress Analysis
5. Optimization & Design Reflection
6. Report & Presentation